

Practical 2 - Non-parametric analysis of survival data

Survival Analysis

London School of Hygiene & Tropical Medicine

Datasets required: `pbcbase`, `whitehall`

R packages required: `survival`, `ggplot2`, `survminer`

```
library(survival)
library(ggplot2)
library(survminer)
```

Introduction

This practical is in two parts. We will use the Primary Biliary Cirrhosis (PBC) data and the Whitehall Study data, which are familiar to you from Practical 1.

Part A. You will use the Primary Biliary Cirrhosis (PBC) data and estimate survival probabilities and compare survival curves between treatment groups.

Part B. You will use the Whitehall data to investigate the effect changing the origin and entry dates has on survival curves.

Aims

The aims of this session are:

- Calculate and interpret estimates of survival probabilities from data with censoring, using the Kaplan-Meier estimator.
- Construct and interpret Kaplan-Meier plots.
- Perform and interpret a log rank test.
- Understand the impact of changing the analysis timescale on survival estimates.

Part A: Primary Biliary Cirrhosis data

The first dataset we will use is called `pbcbase`. This is data from a multicentre, double-blind, clinical trial for the treatment of Primary Biliary Cirrhosis (PBC). There are 184 individuals in the data and 17 variables; the variables of interest in this session are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>datein</code>	Date person entered the study
<code>dateout</code>	Date of the end of follow-up due to either death or censoring
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years
<code>treat</code>	Treatment: 1=placebo, 2=active

Load the data using:

```
pbcbase<-read.table("datasets/pbcbase.csv",sep="," ,header=T)
```

Question 1

We will begin by obtaining Kaplan-Meier estimates of overall survival. In R you can create Kaplan-Meier estimates of survival probabilities $S(t) = \Pr(T > t)$ using:

```
pbcbase.km <- survfit(Surv(time,d)~1,data=pbcbase)
```

An alternative is to use:

```
pbcbase.km <- survfit(Surv(as.numeric(dateout),d,  
                          origin=as.numeric(datein))~1,data=pbcbase)
```

- You can see the Kaplan-Meier estimates of survival probabilities using `summary(pbcbase.km)`. What is the estimated probability of survival beyond 1 year? 5 years?
- Using the Kaplan-Meier estimator for $S(t)$, complete the table below to obtain estimates of $S(t)$ for the first 5 event or censoring times in the data. Check your answers against what is shown in the R output.

Event and censoring times t_j	Number at risk	Number of events	Number of censorings	$\hat{S}(t_j)$
0.0219	184	0	1	
0.0246	183	2	0	
0.0520	181	1	0	
0.1040	180	2	0	
0.1314	178	1	0	

Question 2

- Plot the Kaplan-Meier survival curve using `plot(pbc.km)` and look at how the curve is constructed from the table of times and probabilities. Add labels to the axes (using `xlab=" "` and `ylab=" "`) to make the plot more informative.
- Another way of creating the plot is using `ggsurvplot` in the `survminer` package. Take a look at what information this gives.

```
ggsurvplot(pbc.km, data = pbc, risk.table = T, conf.int = T)
```

Question 3

Next we will describe the survival experience of the two treatment groups using the Kaplan-Meier method.

- Obtain a Kaplan-Meier plot of the estimated survival functions in the two treatment groups using:

```
pbc.km <- survfit(Surv(time,d)~treat,data=pbc)

plot(pbc.km, conf.int=T, col=c("black", "grey"),
     xlab="Time since randomisation (years)", ylab="Survival probability")

legend("topright", c("Placebo", "Active treatment"),
     col=c("black", "grey"))
```

- Summarise survival in the two treatment groups, and the difference in survival.

Question 4

We will use the log rank test to formally compare survival curves in the two treatment groups.

- What is the null hypothesis?
- Perform the test using

```
survdif(Surv(time,d)~treat,data=pbcr)
```

- Interpret the results from the log rank test.

Part B: Whitehall Study

We now return to the Whitehall dataset used in Practical 1. In this practical we will use the variables in the table below.

Variable	Description
id	Unique identifier for each participant
timebth	Date of birth
timein	Date person entered the study
timeout	Date of the end of follow-up due to either death or censoring
chd	Event indicator at the end of follow-up: 0=alive or died from other cause, 1=death due to coronary heart disease
sbpgrp	Systolic blood pressure (SBP) recorded at cohort entry, categorised as: 1: $SBP < 120$; 2: $120 \leq SBP < 140$; 3: $140 \leq SBP < 160$; 4: $SBP \geq 160$

Load the data using:

```
whl<-read.table("datasets/whitehall.csv",sep=";",header=T)
```

Question 1

First we will investigate survival in this cohort using time-in-study as the timescale, i.e. the time origin is the time of joining the cohort.

- Format the date variables as dates, like you did in Practical 1.
- How many deaths were there and what was the total person-time of follow-up, in years.
- Obtain a Kaplan-Meier plot and interpret the results. What is probability of survival beyond 10 years?

Question 2

We will now investigate how changing the time origin and the time period for which an individual is under observation (or ‘at risk’) changes the Kaplan-Meier plot. Obtain Kaplan-Meier curves for the following three different approaches and discuss what you see and how the plots can be interpreted:

- (i) Time origin and start of period in which the participant is ‘at risk’: Date a participant entered the study
- (ii) Time origin and start of period in which the participant is ‘at risk’ = Participant’s date of birth
- (iii) Time origin = Date of birth; start of period in which the participant is ‘at risk’ = date a participant entered the study

Question 3

In this questions we will compare the survival of civil servants who had different levels of systolic blood pressure (SBP) at entry into the study, using the categorical variable `sbpgrp`.

- Obtain Kaplan-Meier estimates of survival in the four SBP groups, and compare.
- Use a log rank test to compare survival across the blood pressure groups. What is the null hypothesis? What are the degrees of freedom for the test? Interpret the results.